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Data Cleaning Documentation Draft

Data Preprocessing

The dataset contains **200,000 records** from 2023 to 2024, with the following columns: **Order_ID**, **Order_Date**, **Customer_Name**, **City**, **State**, **Region**, **Country**, **Category**, **Sub_Category**, **Product_Name**, **Quantity**, **Unit_Price**, **Revenue**, and **Profit**.

The first step was inspecting all column names for inconsistencies. Some columns were found to have **extra spaces** in their names, specifically **Unit_Price**, **Revenue**, and **Profit**. These were renamed to **remove the extra spaces** and ensure clean, consistent column headers.

Next, the data was checked for **blanks** across all columns. Each field was inspected to confirm that **no critical values were missing**, ensuring the dataset was complete before proceeding with analysis.

Following that, the **data types** of each column were **verified and corrected** where necessary. The columns **Quantity**, **Unit_Price**, **Revenue**, and **Profit** were confirmed to be in numeric format since they contain quantitative values. The **Order_Date** column was **converted to a proper date type** to support time-based analysis.

Text fields, including **Customer_Name**, **City**, **State**, **Region**, **Category**, **Sub_Category**, and **Product_Name**, were **trimmed to remove any leading or trailing spaces** that could

cause inconsistencies during filtering and grouping.
Finally, **duplicate records** were identified and **removed** based on the **Order_ID column**, since **Order_ID** serves as a **unique identifier for each transaction**. This step ensured that no order was counted more than once in the analysis.

Data Modeling

The dataset was originally in a flat structure and was transformed into a **star schema** by separating attributes into **dimension tables**. The **original sales table** served as the **fact table, named FactSales**, which **retained** the columns **Order_ID**, **Order_Date**, **Customer_Name**, **Product_Name**, **City**, **State**, **Region**, **Country**, **Quantity**, **Unit_Price**, **Revenue**, and **Profit**.

Four dimension tables were created from the fact table. The **date dimension, DimDate**, was created using DAX and contains the columns **Date**, **Year**, **Quarter**, **Month Number**, **Month Name**, **Week Number**, **Day**, and **Day Name**. This table supports time-based analysis such as monthly trends, quarterly comparisons, and seasonality.

The **customer dimension**, **DimCustomer**, contains **Customer_Name** and was derived from the distinct values in the fact table. The **product dimension, DimProduct**, contains **Product_Name**, **Category**, and **Sub_Category** to support product-level and category-level analysis. The **location dimension**, **DimLocation**, contains **City**, **State**, **Region**, and **Country** to enable geographic filtering and regional comparisons.

Relationships were then established in the **Model view** by connecting **FactSales to each dimension table**. **Order_Date was linked to DimDate**, **Customer_Name to DimCustomer**, **Product_Name to DimProduct**, and **City to DimLocation**. This structure improved filtering, readability, and overall analytical performance of the dashboard.

Metrics

The following key performance indicators (KPIs) were created to summarize overall business performance. **Total Revenue** was calculated by **summing all values in the Revenue column**. **Total Profit** was calculated by **summing all values in the Profit column**.

Total Orders was determined by **counting the distinct values in the Order_ID column** to ensure each order is **counted only once**. **Total Quantity** was calculated by **summing all values in the Quantity column**. **Average Order** Value was computed by **dividing Total Revenue by Total Orders**, representing the average amount generated per transaction.

Profit Margin was calculated by **dividing Total Profit by Total Revenue**, expressing how much profit is earned for every dollar of revenue generated. For time-based analysis, Year-to-Date Revenue and Year-to-Date Profit were also created to track cumulative performance within a given year.

To support trend comparison, additional time-intelligence measures were created. **Prior Revenue Period** was calculated by shifting the **Total Revenue back by one year** using **DATEADD**, allowing comparison of the current revenue against the same period in the previous year.

Revenue Growth Percentage was then derived by **dividing the difference between Total Revenue and Prior Revenue Period** by the **Prior Revenue Period**, expressing how much revenue has grown or declined relative to the prior year. **Prior Period Orders** was calculated by **shifting Total Orders back by one month**, enabling a month-over-month comparison of order volume.